Course Two

Get Started with Python

Instructions

Use this PACE strategy document to record decisions and reflections as you work through this end-of-course project. You can use this document as a guide to consider your responses and reflections at different stages of the data analytical process. Additionally, the PACE strategy documents can be used as a resource when working on future projects.

Course Project Recap

Regardless of which track you have chosen to complete, your goals for this project are:

- [ ] Complete the questions in the Course 2 PACE strategy document
- [x] ~~Answer the questions in the Jupyter notebook project file~~
- [x] ~~Complete coding prep work on project's Jupyter notebook~~
- [x] ~~Summarize the column Dtypes~~
- [x] ~~Communicate important findings in the form of an executive summary~~

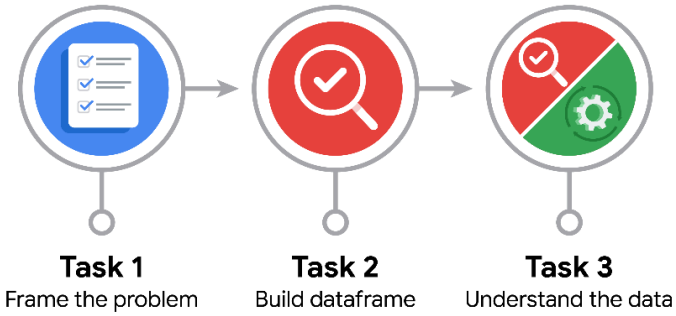
Relevant Interview Questions

Completing the end-of-course project will help you respond these types of questions that are often asked during the interview process:

- Describe the steps you would take to clean and transform an unstructured data set.
- What specific things might you look for as part of your cleaning process?
- What are some of the outliers, anomalies, or unusual things you might look for in the data cleaning process that might impact analyses or ability to create insights?

Reference Guide

This project has three tasks; the visual below identifies how the stages of PACE are incorporated across those tasks.



Data Project Questions & Considerations

PACE: Plan Stage

- How can you best prepare to understand and organize the provided information?

  Read the project background and data dictionary carefully to understand what each column represents. Perform an initial data exploration (EDA) using methods like `.info()`, `.describe()`, and `.head()` to check for data types, missing values, and the general structure of the dataset.

- What follow-along and self-review codebooks will help you perform this work?

  Python libraries' documentation (specifically **Pandas** for data manipulation and **Seaborn/Matplotlib** for visualization) will be essential. Reviewing previous course notebooks on data cleaning and exploratory data analysis will provide a solid template for the workflow.

- What are some additional activities a resourceful learner would perform before starting to code?

  Define the clear business objective (e.g., classifying TikTok claims vs. opinions). Search for industry standards on social media metrics to understand engagement patterns. Outline a step-by-step plan for data cleaning (handling nulls, duplicates) and decide which key variables (like `video_view_count` or `claim_status`) are most relevant to the analysis.

PACE: Analyze Stage

- Will the available information be sufficient to achieve the goal based on your intuition and the analysis of the variables?

Yes. The dataset contains key engagement metrics such as view counts, like counts, and comment counts, along with the `claim_status` label. Based on the initial analysis, there is a clear distinction between claim and opinion videos, which is sufficient to proceed with building a classification model.

- How would you build summary dataframe statistics and assess the min and max range of the data?

I would use the `.describe()` method in Pandas to generate summary statistics (mean, standard deviation, min, max, and quartiles) for all numeric columns. To assess the range specifically, I would look at the `min` and `max` values to identify potential outliers or data entry errors, such as negative view counts or extremely high durations.

- Do the averages of any of the data variables look unusual? Can you describe the interval data?

The averages for "claim" videos are significantly higher than "opinion" videos, particularly in `video_view_count`, which is expected but notable. The interval data (like views and likes) shows a wide spread with a right-skewed distribution, meaning a few videos have extremely high engagement while the majority stay at lower levels. The boxplot visualization confirmed that the interquartile range (IQR) for claim videos is much larger than that of opinions.

PACE: Construct Stage

Note: The Construct stage does not apply to this workflow. The PACE framework can be adapted to fit the specific requirements of any project.

PACE: Execute Stage

- Given your current knowledge of the data, what would you initially recommend to your manager to investigate further prior to performing exploratory data analysis?

  Cevap: I would recommend investigating the data collection process, specifically focusing on how "claims" and "opinions" are tagged. Ensuring the labeling consistency at the source is critical before deep analysis. Additionally, I would suggest checking for potential data leakage or biases in the user engagement metrics.

- What data initially presents as containing anomalies?

  The `video_view_count` and engagement metrics (likes, shares) show significant outliers, especially in "claim" videos. Some videos have millions of views while others have very few, creating a highly skewed distribution. I also identified 2 missing values in the `video_view_count` column that needed to be addressed.

- What additional types of data could strengthen this dataset?

  Adding user-level information such as "account age," "average historical views per user," or "previous ban history" would provide more context for classification. Furthermore, including a "timestamp" for when the video was posted would allow for time-series analysis to see how engagement grows over time for claims versus opinions.